

RESEARCH ARTICLE

Log-Sobolev inequality for near critical Ising models

Roland Bauerschmidt Benoit Dagallier 

DPMMS, University of Cambridge,
Cambridge, UK

Correspondence

Roland Bauerschmidt and Benoit
Dagallier, DPMMS, University of
Cambridge, Cambridge, UK.
Email: rb812@cam.ac.uk and
bd444@cam.ac.uk

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Abstract

For general ferromagnetic Ising models whose coupling matrix has bounded spectral radius, we show that the log-Sobolev constant satisfies a simple bound expressed only in terms of the susceptibility of the model. This bound implies very generally that the log-Sobolev constant is uniform in the system size up to the critical point (including on lattices), without using any mixing conditions. Moreover, if the susceptibility satisfies the mean-field bound as the critical point is approached, our bound implies that the log-Sobolev constant depends polynomially on the distance to the critical point and on the volume. In particular, this applies to the Ising model on subsets of $\mathbb{Z}^d$ when $d > 4$.

The proof uses a general criterion for the log-Sobolev inequality in terms of the Polchinski (renormalisation group) equation, a recently proved remarkable correlation inequality for Ising models with general external fields, the Perron–Frobenius theorem, and the log-Sobolev inequality for product Bernoulli measures.

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1 | MAIN RESULT AND PROOF

1.1 | Main result

Let Λ be a finite set, and denote by $(\cdot, \cdot)$ the standard inner product on $\mathbb{R}^\Lambda$. For $A = (A_{xy})_{x,y \in \Lambda}$ a symmetric matrix with $A_{xy} \leq 0$ if $x \neq y$, the ferromagnetic Ising model with coupling matrix A , inverse temperature $\beta \geq 0$, and external field $h \in \mathbb{R}^\Lambda$ has the expectation

$$\mathbb{E}_{\mu_{\beta,h}} F \propto \sum_{\sigma \in \{\pm 1\}^\Lambda} e^{-\frac{\beta}{2}(\sigma, A \sigma)} e^{(h, \sigma)} F(\sigma). \quad (1.1)$$

We assume that A is positive definite and has spectral radius $\|A\| \leq 1$. The positive definiteness can always be imposed by adding a diagonal matrix (without changing the corresponding Ising model) and $\|A\| \leq 1$ corresponds to a normalisation of the inverse temperature parameter.

For $x \in \Lambda$, let $\sigma^x \in \{\pm 1\}^\Lambda$ be the spin configuration obtained from $\sigma \in \{\pm 1\}^\Lambda$ by flipping the sign of σ_x . The standard (Glauber) Dirichlet form associated with the Ising model (1.1) is

$$D_{\mu_{\beta,h}}(F) = \frac{1}{2} \sum_{x \in \Lambda} \mathbb{E}_{\mu_{\beta,h}} \left[(F(\sigma) - F(\sigma^x))^2 \right]. \quad (1.2)$$

The log-Sobolev constant $\gamma_{\beta,h}$ is the largest constant such that, for all nonnegative F ,

$$\text{Ent}_{\mu_{\beta,h}}(F) \leq \frac{2}{\gamma_{\beta,h}} D_{\mu_{\beta,h}} \left(\sqrt{F} \right), \quad (1.3)$$

where $\text{Ent}_{\mu_{\beta,h}}(F) = \mathbb{E}_{\mu_{\beta,h}} \Phi(F) - \Phi(\mathbb{E}_{\mu_{\beta,h}} F)$ with $\Phi(x) = x \log x$ is the relative entropy. Our main result is stated in terms of the 0-field susceptibility of the Ising model, given by

$$\chi_\beta = \sup_{x \in \Lambda} \sum_{y \in \Lambda} \mathbb{E}_{\beta,0}(\sigma_x \sigma_y). \quad (1.4)$$

Theorem 1.1. *The log-Sobolev constant of (1.1) satisfies*

$$\frac{1}{\gamma_{\beta,h}} \leq \frac{1}{2} + \int_0^\beta e^{2 \int_0^t \chi_s ds} dt. \quad (1.5)$$

The Dirichlet form is associated with a Markov process (Glauber dynamics), see [26] for background. The standard Dirichlet form (1.2) corresponds to a convenient choice of transition rates, but other common choices of rates (such as heat-bath and Metropolis dynamics) are equivalent when the rates are bounded, as is the case for the standard Ising model on $\Lambda \subset \mathbb{Z}^d$, also see [26]. The log-Sobolev inequality can be equivalently formulated in terms of hypercontractivity of the Glauber semigroup and has many well-known implications. In particular, the inverse log-Sobolev constant bounds the relaxation rate to equilibrium in the sense of relative entropy, that is, if F_t $\mu_{\beta,h}$ denotes the law of the dynamics at time $t \geq 0$, then

$$\text{Ent}_{\mu_{\beta,h}}(F_t) \leq e^{-2\gamma_{\beta,h}t} \text{Ent}_{\mu_{\beta,h}}(F_0). \quad (1.6)$$

The inverse log-Sobolev constant can also be used to obtain bounds on other notions of convergence to equilibrium in finite and infinite volume, see the references [20, 25–27] for introductions to the general implications.

Since χ_s is increasing in s (by the second Griffiths inequality), Theorem 1.1 shows in particular that whenever χ_β is bounded, the log-Sobolev constant is bounded below:

$$\frac{1}{\gamma_{\beta,h}} \leq \frac{1}{2} + \beta e^{2\beta\chi_\beta}. \quad (1.7)$$

This shows that the log-Sobolev inequality holds uniformly in the volume in the entire high temperature phase of the Ising model, for example, on $\Lambda \uparrow \mathbb{Z}^d$ but equally in a much more general setting. For lattices Λ , uniformity in the volume was recently proved in [15] by establishing the strong spatial mixing property, see [26]. Our more general criterion does not rely on geometry or mixing conditions. The new correlation inequality from [15] is also important in our proof.

But even more interestingly, the bound (1.5) shows that if the susceptibility satisfies the mean-field bound then the log-Sobolev constant is polynomial (in the distance to the critical point and in the volume). This is summarised in the following corollary, whose proof simply consists of carrying out the integrals in (1.5) using the assumed bound on χ_β .

Corollary 1.2. *Let $D > 1/2$ and assume that the mean-field bound $\chi_\beta \leq D/(\beta_c - \beta)$ for all $\beta < \beta_c$ holds. Then*

$$\frac{1}{\gamma_{\beta,h}} \leq \frac{1}{2} + \frac{\beta_c}{2D-1} [(1 - \beta/\beta_c)^{1-2D} - 1] \underset{\beta \rightarrow \beta_c}{\sim} \frac{\beta_c}{2D-1} (1 - \beta/\beta_c)^{1-2D}. \quad (1.8)$$

Similarly, if the finite-volume mean-field bound $\chi_\beta \leq D/(\beta_c - \beta + L^{-2})$ for all $\beta < \beta_c$ holds, then

$$\frac{1}{\gamma_{\beta_c,h}} \leq \frac{1}{2} + \frac{\beta_c + L^{-2}}{2D-1} [(L^2\beta_c + 1)^{2D-1} - 1] \underset{L \rightarrow \infty}{\sim} \frac{\beta_c}{2D-1} (L^2\beta_c + 1)^{2D-1}. \quad (1.9)$$

The mean-field bound holds for Ising models on $\mathbb{Z}^d$ in $d \geq 5$, see [1, 2], and the finite-volume mean-field bound similarly holds for the Ising model on a hypercube of side length L (with free boundary conditions) in $d \geq 5$, and with possibly a different exponent for other boundary conditions, see [12] for discussion of this. The exponents above are not the expected optimal ones when $\Lambda \uparrow \mathbb{Z}^d$, which would be -1 respectively $+1$ in $d \geq 5$. We remark though that in the actual mean-field model (the Curie–Weiss model), the mean-field bound holds with $D = 1$ (see, for example [8, Chapter 1]) and one can thus obtain the optimal exponent from (1.5). However, for the Curie–Weiss model, also many other methods give sharp results, see, for example [26, 28], and the more recent spectral conditions [3, 5, 17].

For lattices, by contrast, a polynomial bound on the spectral gap for the Glauber dynamics of the Ising model at the critical temperature has previously only been proved in two dimensions [24]. The proof of this bound relies on the RSW estimates [16] as input. Similar estimates have also been obtained for two-dimensional q -state Potts models with $q < 4$, see [18]. The approach of the critical point for these models appears to remain open. Other polynomial bounds for near critical spin models were restricted to simpler geometries such as trees [14] or hierarchical lattices [6].

For completeness, we also comment briefly on the low-temperature phase, in which our technique to prove Theorem 1.1 does not seem directly useful. In this region, the Glauber dynamics equilibrates at a rate that typically vanishes with the size of Λ (assuming the external field h is 0). For the ferromagnetic Ising model in a hypercube $\Lambda \subset \mathbb{Z}^d$ of side length L , boundary conditions play a crucial role, and their influence is not captured by the zero-field susceptibility (1.4) (zero field corresponds to free boundary conditions). For instance, for free boundary conditions, it is known that the spectral gap (which is always larger than the log-Sobolev constant $\gamma_{\beta,h}$) decays like $e^{-c(\beta)L^{d-1}(1+o_L(1))}$ when $\beta > \beta_c$; the constant $c(\beta)$ is explicit in dimension two [26] and there are more general bounds in dimension three and higher [10]. Conversely, if the boundary condition is all +, there is a long-standing conjecture, the so-called Lifshitz law, that convergence to equilibrium from a worst-case initial condition should typically take a time of order L^2 , up to logarithmic corrections. Arguments supporting that bound are given in [11], but so far the bound has only been established at 0 temperature in all dimensions $d \geq 2$, see [13] in dimension $d = 2, 3$ and [21] in higher dimension. For $\beta \in (\beta_c, \infty)$, the currently best known results are in dimension two, where the rate of relaxation is known to be going to 0 at least as $L^{-c'(\beta)\log L}$ for some $c'(\beta) > 0$, see [23] and references therein. Similar bounds can also be obtained with periodic boundary conditions instead of all +, but for well chosen initialisations of the dynamics: bounds on the mixing time of order L^{-1} in dimension $d = 2$ and $L^{c'(\beta)\log^{d-1} L}$ in higher dimensions have recently been established in [19].

In [9], we derive a variant of Theorem 1.1 for φ^4 models and use it to prove that the continuum φ_2^4 and φ_3^4 measures satisfy log-Sobolev inequalities uniformly in the regularisation (and volume in the uniqueness phase).

Remark 1.3. The proof only uses that χ_β defined in (1.4) provides an upper bound on the spectral radius of the two-point function $(\mathbb{E}_{\mu_{\beta,0}}(\sigma_x \sigma_y))_{x,y \in \Lambda}$. Thus replacing χ_β by the spectral radius of the two-point function gives a slightly more general statement.

1.2 | Proof of Theorem 1.1

The strategy of the proof is to decompose $\mu_{\beta,h}$ into two measures: an infinite-temperature Ising part, for which the log-Sobolev inequality is known; and a continuous part for which the log-Sobolev inequality is established via the criterion based on the Polchinski (renormalisation group) equation from [7, Theorem 2.5]. The first part is similar to the one in [5] where the same kind of decomposition was performed. In [5], the second part was treated with the Bakry–Emery criterion, leading to the condition $\beta < 1$ (but without assuming the coupling matrix A is ferromagnetic). In contrast, our use of the more powerful Polchinski equation criterion does not require an upper bound on β . To verify the assumptions of the criterion we exploit that A is ferromagnetic by using the FKG inequality and the new correlation inequality from [15].

Proof of Theorem 1.1. By taking a limit at the end, we can assume $\|A\| < 1$ so that all covariances below are nondegenerate. Fix $\alpha > \beta$ and define for $t \in [0, \beta]$:

$$C_t = (tA + (\alpha - t))^{-1}. \quad (1.10)$$

The covariances C_t are positive definite, increasing as quadratic forms in $t \in [0, \beta]$, and satisfy

$$\dot{C}_t = (1 - A)C_t^2, \quad \ddot{C}_t = 2(1 - A)C_t \dot{C}_t. \quad (1.11)$$

The Ising measure $\mu_{\beta, h}$ can then be written as (the term $\alpha - t = \alpha - \beta$ is irrelevant since $\sigma_x^2 = 1$)

$$\mathbb{E}_{\mu_{\beta, h}} F \propto \sum_{\sigma \in \{\pm 1\}^\Lambda} e^{-\frac{1}{2}(\sigma, C_\beta^{-1} \sigma)} e^{(h, \sigma)} F(\sigma) \propto \sum_{\sigma \in \{\pm 1\}^\Lambda} e^{-\frac{\beta}{2}(\sigma, A \sigma)} e^{(h, \sigma)} F(\sigma). \quad (1.12)$$

For $t \in [0, \beta)$ define the renormalised potential for $\varphi \in \mathbb{R}^\Lambda$ by

$$\begin{aligned} V_t(\varphi) &= -\log \sum_{\sigma \in \{\pm 1\}^\Lambda} e^{-\frac{1}{2}(\sigma - \varphi, C_t^{-1}(\sigma - \varphi))} e^{(h, \sigma)} \\ &= \frac{1}{2}(\varphi, C_t^{-1} \varphi) - \log \sum_{\sigma \in \{\pm 1\}^\Lambda} e^{-\frac{1}{2}(\sigma, C_t^{-1} \sigma)} e^{(h + C_t^{-1} \varphi, \sigma)}, \end{aligned} \quad (1.13)$$

and the renormalised measure as the probability measure on $\mathbb{R}^\Lambda$ given by

$$\nu_{t, \beta}(d\varphi) \propto e^{-\frac{1}{2}(\varphi, (C_\beta - C_t)^{-1} \varphi)} e^{-V_t(\varphi)} d\varphi. \quad (1.14)$$

Both V_t and $\nu_{t, \beta}$ also depend on h (not denoted explicitly). For any $t \in (0, \beta]$ then

$$e^{-V_t(\varphi)} \propto \int_{\mathbb{R}^\Lambda} e^{-\frac{1}{2}(\zeta, (C_t - C_0)^{-1} \zeta)} e^{-V_0(\varphi + \zeta)} d\zeta, \quad (1.15)$$

and the Ising measure $\mu_{\beta, h}$ can be decomposed as follows for any $t \in [0, \beta)$ (thus including $t = 0$):

$$\mathbb{E}_{\mu_{\beta, h}} F = \mathbb{E}_{\nu_{t, \beta}} \mathbb{E}_{\mu_{t, h + C_t^{-1} \varphi}} F. \quad (1.16)$$

The last two equations follow from the following convolution identity for Gaussian densities, valid for all $0 \leq t < s \leq \beta$ and $\sigma \in \mathbb{R}^\Lambda$:

$$e^{-\frac{1}{2}(\sigma, C_s^{-1} \sigma)} \propto \int_{\mathbb{R}^\Lambda} e^{-\frac{1}{2}(\varphi, (C_s - C_t)^{-1} \varphi)} e^{-\frac{1}{2}((\sigma - \varphi), C_t^{-1}(\sigma - \varphi))} d\varphi. \quad (1.17)$$

Let $F : \{-1, 1\}^\Lambda \rightarrow \mathbb{R}_+$ be fixed. The relative entropy can be decomposed using (1.16):

$$\text{Ent}_{\mu_{\beta, h}}(F) = \mathbb{E}_{\nu_{t, \beta}} \left(\text{Ent}_{\mu_{t, h + C_t^{-1} \varphi}}(F(\sigma)) \right) + \text{Ent}_{\nu_{t, \beta}}(G(\varphi)), \quad (1.18)$$

where

$$G(\varphi) = \mathbb{E}_{\mu_{t, h + C_t^{-1} \varphi}} F(\sigma). \quad (1.19)$$

In fact, we only use $t = 0$ in this identity. Then $C_0^{-1} = \alpha$ and the measure $\mu_{0, h + C_0^{-1} \varphi}$ is the infinite temperature (product) Ising model with external field $h + \alpha \varphi$. It thus satisfies a log-Sobolev

inequality uniformly in h and φ with log-Sobolev constant $2/\gamma = 1$, see [4, 22, 27]. Therefore

$$\mathbb{E}_{\nu_{0,\beta}} \left(\text{Ent}_{\mu_{0,h+\alpha\varphi}}(F(\sigma)) \right) \leq \mathbb{E}_{\nu_{0,\beta}} D_{\mu_{0,h+\alpha\varphi}}(\sqrt{F}) = D_{\mu_{\beta,h}}(\sqrt{F}). \quad (1.20)$$

For the second term in (1.18), we show that $\nu_{0,\beta}$ satisfies a log-Sobolev inequality (with constant denoted by γ) by using the Polchinski equation criterion from [7, Theorem 2.5]. This criterion states that, for a family $(\dot{\lambda}_t)_{t \in [0,\beta]}$ of real numbers and increasing covariances C_t as above:

$$\forall t \in [0, \beta], \quad \dot{C}_t \text{Hess } V_t(\varphi) \dot{C}_t - \frac{1}{2} \ddot{C}_t \geq \dot{\lambda}_t \dot{C}_t \quad (1.21)$$

implies

$$\frac{1}{\gamma} \leq \|\dot{C}_0\| \int_0^\beta e^{-2 \int_0^t \dot{\lambda}_s ds} dt. \quad (1.22)$$

To be precise, in [7, Theorem 2.5], the covariances C_t are parametrised by $t \in [0, \infty]$ rather than $t \in [0, \beta]$, and start at $C_0 = 0$. This difference simply corresponds to the change of variable from t to $\beta(1 - e^{-t})$ in the definition of C_t , and to replacing C_t by $C_t - C_0$. We prefer the parametrisation by $t \in [0, \beta]$ here due to its more transparent interpretation as a change in inverse temperature.

To find a family $(\dot{\lambda}_t)_{t \in [0,\beta]}$, note that (1.13) implies

$$\text{Hess } V_t(\varphi) = C_t^{-1} - C_t^{-1} \Sigma_t (h + C_t^{-1} \varphi) C_t^{-1} \quad (1.23)$$

where

$$\Sigma_t(f) = \left(\mathbb{E}_{\mu_{t,f}}(\sigma_x \sigma_y) - \mathbb{E}_{\mu_{t,f}}(\sigma_x) \mathbb{E}_{\mu_{t,f}}(\sigma_y) \right)_{x,y}, \quad f \in \mathbb{R}^\Lambda. \quad (1.24)$$

Since $\mu_{t,f}$ is a ferromagnetic Ising model (at inverse temperature t and with external field $f \in \mathbb{R}^\Lambda$), the FKG inequality implies that $\Sigma_t(f)$ has nonnegative entries, and the recently proved powerful correlation inequality [15, Corollary 1.3] further shows that $\Sigma_t(f)_{xy} \leq \Sigma_t(0)_{xy}$ for all $x, y \in \Lambda$. Finally, by the Perron–Frobenius theorem, there exists $Y_0 = Y_0(f) \in \mathbb{R}^\Lambda$ with nonnegative entries and $\|Y_0\|_2 = 1$ such that

$$\|\Sigma_t(f)\| = Y_0^T \Sigma_t(f) Y_0 \leq Y_0^T \Sigma_t(0) Y_0 \leq \|\Sigma_t(0)\| \leq \sup_{x \in \Lambda} \sum_{y \in \Lambda} \Sigma_t(0)_{xy} = \chi_t,$$

where $\|\Sigma\| = \sup_{\|Y\|_2 \leq 1} \|\Sigma Y\|_2$ is the spectral radius of the symmetric matrix Σ , and the last inequality follows from the Cauchy-Schwarz inequality. Thus for all $X \in \mathbb{R}^\Lambda$, with $f = h + C_t^{-1} \varphi$,

$$X^T C_t^{-1} \Sigma_t (h + C_t^{-1} \varphi) C_t^{-1} X \leq \|C_t^{-1} X\|_2^2 \|\Sigma_t(h + C_t^{-1} \varphi)\| \leq \chi_t X^T C_t^{-2} X, \quad (1.26)$$

and we have the quadratic form inequality ($C_t, \dot{C}_t, \ddot{C}_t$ and A all commute)

$$\begin{aligned} \dot{C}_t \text{Hess } V_t(\varphi) \dot{C}_t - \frac{1}{2} \ddot{C}_t &\geq \dot{C}_t^2 C_t^{-1} - \chi_t \dot{C}_t^2 C_t^{-2} - (1 - A) C_t \dot{C}_t \\ &= -(1 - A) \dot{C}_t \chi_t \geq -\chi_t \dot{C}_t. \end{aligned} \quad (1.27)$$

Therefore by (1.22),

$$\text{Ent}_{\nu_{0,\beta}}(G) \leq \frac{2}{\gamma} \mathbb{E}_{\nu_{0,\beta}} (\nabla \sqrt{G})^2 \quad (1.28)$$

with

$$\frac{1}{\gamma} \leq \|\dot{C}_0\| \int_0^\beta e^{2 \int_0^t \chi_s ds} dt = \frac{1}{\alpha^2} \int_0^\beta e^{2 \int_0^t \chi_s ds} dt. \quad (1.29)$$

To conclude, it only remains to bound $(\nabla_\varphi \sqrt{G})^2$ in terms of F . This argument is exactly as in [5]. Indeed, by the definitions of G (with $t = 0$) and of $\mu_{0,h+\alpha\varphi}$,

$$\forall x \in \Lambda, \quad (\nabla_{\varphi_x} \sqrt{G})^2 = \alpha^2 \frac{\text{Cov}_{\mu_{0,h+\alpha\varphi}}(F, \sigma_x)^2}{4 \left(\mathbb{E}_{\mu_{0,h+\alpha\varphi}} F \right)}. \quad (1.30)$$

Denote by $\mu_{0,h+\alpha\varphi}^x$ the conditional measure of the product measure $\mu_{0,h+\alpha\varphi}$ with all σ_y with $y \neq x$ fixed. By duplication, the covariance can be written as

$$\begin{aligned} \text{Cov}_{\mu_{0,h+\alpha\varphi}^x}(F, \sigma_x)^2 &= \left(\frac{1}{2} \mathbb{E}_{\mu_{0,h+\alpha\varphi}^x} \otimes \mu_{0,h+\alpha\varphi}^x \left[(\sqrt{F} - \sqrt{F'}) (\sqrt{F} + \sqrt{F'}) (\sigma_x - \sigma'_x) \right] \right)^2 \\ &\leq \text{Var}_{\mu_{0,h+\alpha\varphi}^x}(\sqrt{F}) \frac{1}{2} \mathbb{E}_{\mu_{0,h+\alpha\varphi}^x} \otimes \mu_{0,h+\alpha\varphi}^x \left[(\sqrt{F} + \sqrt{F'})^2 (\sigma_x - \sigma'_x)^2 \right] \\ &\leq 8 \text{Var}_{\mu_{0,h+\alpha\varphi}^x}(\sqrt{F}) \mathbb{E}_{\mu_{0,h+\alpha\varphi}^x}(F), \end{aligned} \quad (1.31)$$

where we have used the Cauchy-Schwarz inequality and that $|\sigma_x| \leq 1$. Since $\mu_{0,h+\alpha\varphi}$ is a product measure, $\text{Cov}_{\mu_{0,h+\alpha\varphi}}(F, \sigma_x) = \mathbb{E}_{\mu_{0,h+\alpha\varphi}} \text{Cov}_{\mu_{0,h+\alpha\varphi}^x}(F, \sigma_x)$. Thus, using the Cauchy-Schwarz inequality on the third line:

$$\begin{aligned} \text{Cov}_{\mu_{0,h+\alpha\varphi}}(F, \sigma_x)^2 &= \mathbb{E}_{\mu_{0,h+\alpha\varphi}} \left[\text{Cov}_{\mu_{0,h+\alpha\varphi}^x}(F, \sigma_x) \right]^2 \\ &\leq 8 \mathbb{E}_{\mu_{0,h+\alpha\varphi}} \left[\sqrt{\text{Var}_{\mu_{0,h+\alpha\varphi}^x}(\sqrt{F})} \sqrt{\mathbb{E}_{\mu_{0,h+\alpha\varphi}^x}(F)} \right]^2 \\ &\leq 8 \mathbb{E}_{\mu_{0,h+\alpha\varphi}} \left[\text{Var}_{\mu_{0,h+\alpha\varphi}^x}(\sqrt{F}) \right] \mathbb{E}_{\mu_{0,h+\alpha\varphi}}(F). \end{aligned} \quad (1.32)$$

Injecting the last bound into (1.30) and using the elementary fact that the biased Bernoulli ± 1 measure $\mu_{0,h+\alpha\varphi}^x$ has spectral gap at least 2 (with our normalisation of Dirichlet form),

$$\begin{aligned} \mathbb{E}_{\nu_{0,\beta}} (\nabla_{\varphi_x} \sqrt{G})^2 &\leq 2\alpha^2 \mathbb{E}_{\nu_{0,\beta}} \mathbb{E}_{\mu_{0,h+\alpha\varphi}} \left[\text{Var}_{\mu_{0,h+\alpha\varphi}^x}(\sqrt{F}) \right] \\ &\leq 2\alpha^2 \mathbb{E}_{\nu_{0,\beta}} \mathbb{E}_{\mu_{0,h+\alpha\varphi}} \frac{1}{4} \mathbb{E}_{\mu_{0,h+\alpha\varphi}^x} \left[\left(\sqrt{F(\sigma)} - \sqrt{F(\sigma^x)} \right)^2 \right]. \end{aligned} \quad (1.33)$$

Taking the sum over x and using that $\mathbb{E}_{\mu_{0,h+\alpha\varphi}} \mathbb{E}_{\mu_{0,h+\alpha\varphi}^x} = \mathbb{E}_{\mu_{0,h+\alpha\varphi}}$, therefore

$$\mathbb{E}_{\nu_{0,\beta}} (\nabla_{\varphi} \sqrt{G})^2 \leq \alpha^2 D_{\mu_{\beta,h}}(\sqrt{F}). \quad (1.34)$$

In summary, putting together the entropy decomposition (1.18) and the bounds (1.20) and (1.28)–(1.34) on each term of this decomposition:

$$\text{Ent}_{\mu_{\beta,h}}(F) \leq \left(1 + \frac{2\alpha^2}{\gamma}\right) D_{\mu_{\beta,h}}(\sqrt{F}). \quad (1.35)$$

Substituting (1.29) into this bound gives the result. $\square$

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ORCID

Roland Bauerschmidt  <https://orcid.org/0000-0001-7453-2737>

Benoit Dagallier  <https://orcid.org/0009-0004-5427-5664>

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